Neelan Chung

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Skills

Software & Programming: C/C++, Python, Java, MATLAB, Git, Unix, STM32CubeMX/IDE
Electrical: Circuit Design (Altium, LTspice), Hardware Commmunication Protocols(CAN, I2C, UART, SPI, SWD)
Hardware & Design: CAD (SolidWorks, BricsCAD), Design Reveiw (Navisworks, Intergraph Smart Reveiw)

Education

University of British Columbia — Expected May 2030
B.A.Sc. Integrated Engineering - GPA: 4.33 — *Vancouver, BC*

Experience

Mechanical & Materials Engineering Intern — Aug 2026 – Present
Fluor Canada — *Vancouver, BC*

- 3D modelled **mining infrastructure and bulk-material flows** in **SolidWorks** for large-scale mining projects
- Managed **vendor package documentation**, including signature routing and submitting documentation to Project Management for review, and developing a **VBA automation tool** to rename, organize, and designate vendor files
- Assisted with **gyratory crusher and jib crane equipment packages**, supporting package engineers and the material handling team with technical coordination and documentation

Telemetry Team Lead & Firmware Member — Sept 2025 – Present
UBC ThunderBikes — *Vancouver, BC*

- Developed **embedded systems** using **STM32** microcontrollers, STM32CubeMX, and the Makefile build tool
- Currently designing and writing the firmware for a **telemetry PCB using Altium Designer** to record bike metrics
- Wrote Firmware for a **CAN-based charger monitoring module** to display battery charging values over time
- Assisted with CAD modelling and prototyping for electronic enclosures and charging-port housings
- Collaborated with firmware, electrical, and mechanical sub-teams on system integration and prototyping

Projects

Smart Navigation Goggle Module — *2nd-Year IGEN Capstone* — Jan 2026 – Apr 2026

- Developed object-oriented embedded firmware in **C++** for an **ESP32-S3** wearable Heads-Up Display (HUD) using custom consumer methods, display driver classes, and handling structures with the Arduino framework and IDE
- Implemented **BLE** communication architecture using **GATT** services and characteristics for real-time transmission of navigation and performance data from an Android app that took navigation data from the Google Maps API
- Built data-processing classes to render live pace, distance, timing, and directional metrics on an **OLED** display
- Developed firmware using **ESP32 BLE APIs** and validated stable wireless communication and timed updates
- Contributed to a multidisciplinary project spanning embedded systems, optics, CAD, and wearable hardware prototyping

File Renaming and Organization Program — *Fluor Mining & Metals* — Sept 2026

- Created a file-renaming program in VBA to rename vendor package files using vendor-specified metadata.
- Used UserForms to create a streamlined UI that instructed the user on how to create the project folder structure before feeding it into the program, and allowed the user to specify metadata spreadsheet rows and columns during runtime
- Used VBA UserForms, file system objects, dictionaries, and error handling methods to create a renaming manifest, allowing the user to input missing information, customize file names, and correct potential renaming errors
- Used a modular, segmented approach to developing the program, carefully tracking and managing object scope

CAN-Controller Charger Voltage Display — *UBC ThunderBikes* — Sept 2025

- Used STM32CubeMX to program an **STM32 Blue Pill** to receive battery charging data, including cell voltage and current, from the bike's **CAN bus** and display it on an OLED screen in a metric-over-time format
- Collaborated with the BMS team on firmware development and designed a custom housing using **SolidWorks**

Mining Infrastructure CAD — *Fluor Mining & Metals* — Aug 2026 - Sept 2026

- Designed and modeled dump-pocket and surge-pocket infrastructure in **SolidWorks** for Fluor's El Abra Mine Project, and used surface modeling, feature intersections, and constraints to estimate material volume and fill line locations
- Developed flexible, equation-driven bulk-material fill geometry with multiple configurations to preserve design intent